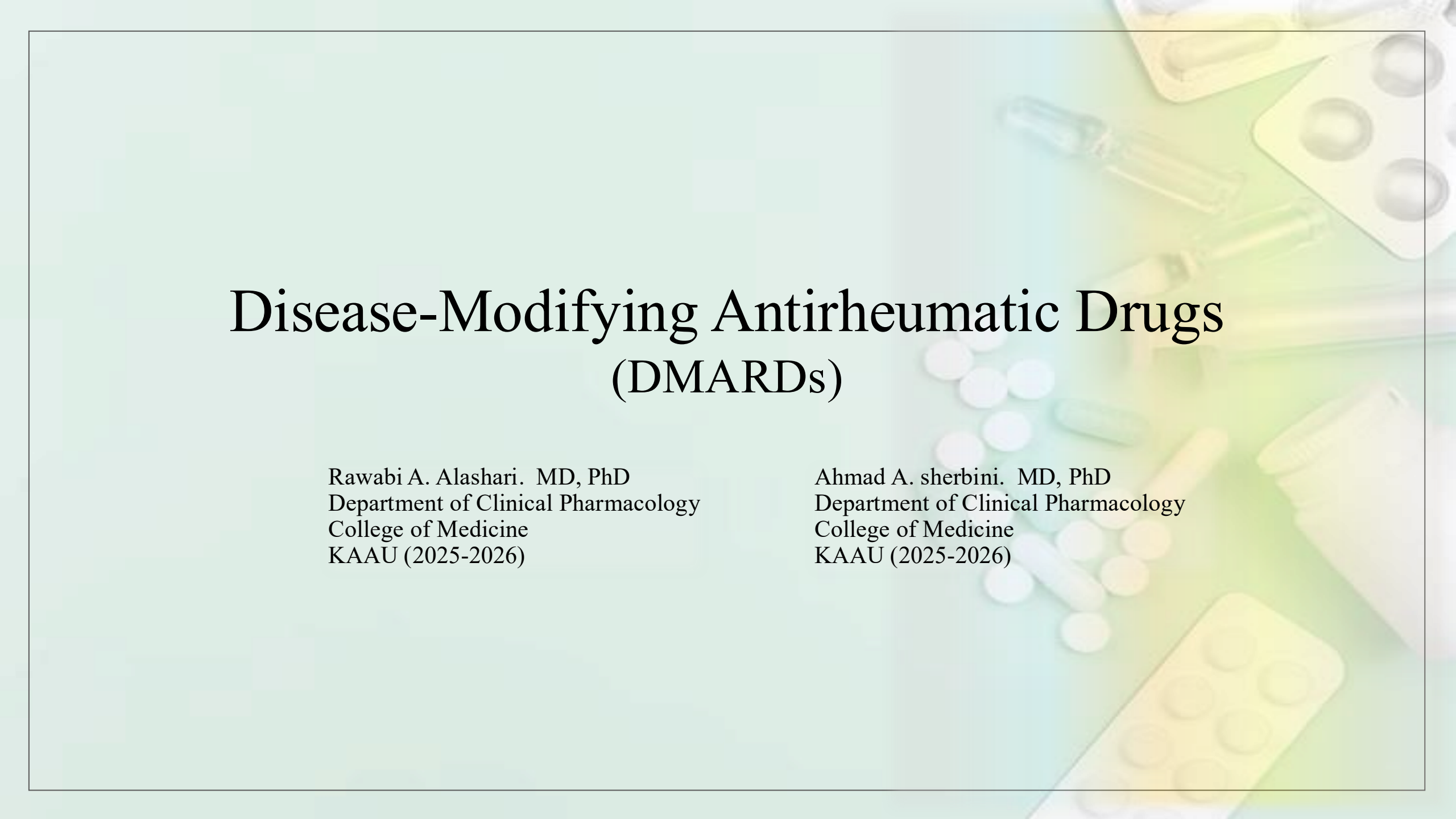




Disease-Modifying Antirheumatic Drugs (DMARDs)

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Learning Objectives

By the end of this lecture, you should be able to:

1. **Define** DMARDs and explain their **role** in inflammatory arthritis
2. **Classify** conventional, biologic, and targeted synthetic DMARDs
3. Explain the **mechanisms of action** of **major** DMARDs
4. Identify **important adverse effects, contraindications, and monitoring parameters**
5. **Compare** DMARDs with NSAIDs and corticosteroids
6. **Apply** this knowledge to common clinical scenarios

Clinical Problem

A 42-year-old woman with rheumatoid arthritis, complains about joint pain and swelling.

She was prescribed NSAIDs which made her feel better. However, 2 years later, she developed joint deformities.

Why did she develop joint deformities despite NSAIDs treatment?

- NSAIDs relieve pain and stiffness
- NSAIDs do **NOT** prevent joint destruction
- Therefore, a disease-modifying drug is needed

This is why DMARDs are the cornerstone of rheumatoid arthritis (RA) management.



<https://doi.org/10.1016/B978-0-12-801238-3.11450-3>



A patient may feel better on NSAIDs
BUT still develop joint damage!!!!



Many of **DMARDs** were developed to treat RA but they are now used to treat a variety of inflammatory disorders



DMARDs slow disease progression



NSAIDs: symptomatic relief only
Steroids: rapid anti-inflammatory effect
DMARDs: long-term disease control



ONLY DMARDs prevent irreversible joint damage



Inflammatory Diseases (Overview)

Why inflammation matters specifically in RA ? How it connects to DMARDs ?

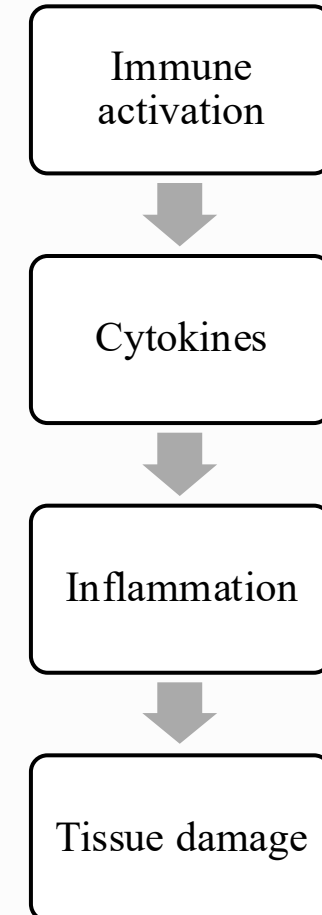
- Key mediators relevant to RA: TNF- α , IL-6, IL-1
- Persistent cytokine-driven inflammation leads to synovial damage, cartilage destruction, and bone erosion
- DMARDs work by targeting these inflammatory pathways:

Treatment:

There are **two** approaches:

1. Symptoms control	NSAIDs, Corticosteroids
2. Disease control	Conventional DMARDs, Biologic DMARDs, Targeted synthetic DMARDs

<https://www.hss.edu/health-library/conditions-and-treatments/dmards-for-rheumatic-disease>



DMARDs (Definition, Classification)

What do DMARDs do ?

1. **Reduce** inflammation
2. **Prevent** (or slow) structural damage
3. **Improve** long-term outcomes

DMARDs act slowly but modify disease progression
Used early to prevent irreversible damage

Classification of DMARDs

Conventional DMARDs	Biologic DMARDs	Targeted synthetic DMARDs
 Methotrexate	TNF inhibitors	AK inhibitors (Tofacitinib, Baricitinib)
Sulfasalazine	IL-6 inhibitor	
Hydroxychloroquine	T-cell blocker	
Leflunomide	B-cell depletion	



Which drug do we start FIRST in RA?

CONVENTIONAL DMARDs

Main traditional agents used in RA

Conventional DMARDs

Methotrexate

Sulfasalazine

Hydroxychloroquine

Leflunomide

**METHOTREXATE:
FIRST-LINE ANCHOR DRUG**



ANCHOR DRUG:

THE FOUNDATIONAL, FIRST-LINE
THERAPY AROUND WHICH ALL
OTHER TREATMENTS ARE BUILT,
MANAGED, AND COMPARED

Methotrexate (The anchor drug)

Methotrexate (MTX)

1. **First-line** DMARDs in rheumatoid arthritis (RA) - unless contraindicated
2. **Once weekly** dosing only(daily dosing result in toxicity)
3. **Have** Anti-inflammatory + immunosuppressive effects
4. **Response** to methotrexate usually occurs **within 3 to 6 weeks** of starting treatment.

Mechanism Of Action (MOA)

Inhibits folate metabolism (MTX is a folic acid antagonist)

1. DNA Synthesis	↓ thymidine + ↓ purines → antiproliferative effect
2. Methylation Cycle	↑ homocysteine (contribute to toxicity)
3. Adenosine Pathway	Folate antagonist → ↓ DNA synthesis → ↑ adenosine (anti-inflammatory mediator)

Effect	Clinical Meaning
↓ Immune activation	Controls disease activity
↓ Inflammation	Reduces pain & swelling
↓ Joint damage	Prevents deformity

- ❖ MTX anti-inflammatory effect is largely linked to increased adenosine
- ❖ Folate antagonism explains MTX toxicity (Folate deficiency → toxicity)

Methotrexate (What really matters clinically)

1. Anti-inflammatory effect (MOST IMPORTANT)

MTX works in RA because it increase the levels of **Adenosine** (an anti-inflammatory mediator)

↑ Adenosine → ↓ inflammation

2. Anti-proliferative effect

Impairs nucleic acid synthesis, thus suppresses rapidly dividing cells

↓ DNA synthesis → ↓ immune cell proliferation (immunosuppressive)

3. Toxicity mechanism

Blocks DHFR. folate can't be activated, shuts down this entire cycle →
↓ cell proliferation (that's why we give folic acid for patients on MTX)

Folate inhibition → side effects

- Folic acid reduces toxicity but does **NOT** reduce efficacy

CLICK HERE



Armando Hasudungan

2.86M subscribers

<https://www.youtube.com/watch?v=2k4-NuFjto0&t=7s>



First-line in RA
MOA: inhibits Dihydro-Folate Reductase (DHFR)



Methotrexate (Clinical use and safety)

Uses:

- Rheumatoid arthritis
- Psoriatic arthritis
- Other inflammatory diseases

Adverse effects:

- GI: nausea, vomiting, anorexia
- Mouth ulcers
- Elevated liver enzymes
- Cytopenias
- Teratogenicity (contraindicated in pregnancy)

Monthly Labs Monitoring:

- Complete Blood Count (CBC)
- Liver Functions Tests (LFTs)
- Renal Functions (Creatinine)

Monitoring patients on MTX is critical to detect potential toxicities:

Labs	Purpose	Risk
CBC	For myelosuppression (bone marrow suppression)	Causes a dangerous drop in white blood cells (infection risk), platelets (bleeding risk), and red blood cells (anemia)
LFTs	For hepatotoxicity (liver damage)	Elevated liver enzymes (ALT, AST) which may indicate acute toxicity or chronic liver fibrosis.
Cr	For kidney health and ability to excrete the drug	Methotrexate is primarily cleared by the kidneys (60%-95%). If kidney function declines, methotrexate accumulates, leading to higher toxicity risk

- ❖ Toxicity risk increases with renal impairment because methotrexate is primarily renally cleared
- ❖ Folic acid supplementation is mandatory
- ❖ MTX can inhibit the fetal dihydrofolate reductase and interfere with fetal DNA synthesis

Other Conventional DMARDs

Comparing them to methotrexate: used when MTX not tolerated or in combination

Conventional DMARDs	Hydroxychloroquine	Sulfasalazine	Leflunomide
MOA	Interferes with antigen presentation and immune signaling	Anti-inflammatory + immunomodulatory	Inhibits pyrimidine synthesis → ↓ lymphocyte proliferation
Main use	- Mild RA - SLE	- RA - Inflammatory bowel disease (IBD)	Contraindicated in pregnancy
Key toxicity	Retinal toxicity	- GI upset - Bone marrow suppression - Hypersensitivity	- Hepatotoxicity - Teratogenicity - Hypertension
Key monitoring	Eye exams	Avoid pregnancy (during and a period after stopping the drug)	- LFTs - Avoid pregnancy



What if methotrexate fails?

BIOLOGICAL DMARDs

Target specific cytokines or immune cells involved in RA

Used when conventional DMARDs fail

Biologic DMARDs
TNF inhibitors
IL-6 inhibitor
T-cell blocker
B-cell depletion

When are biologics used?

After failure of methotrexate +/- combination DMARDs:

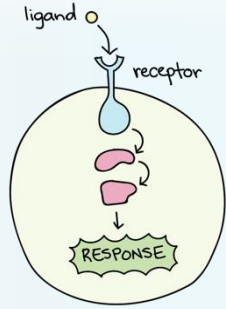
- Inadequate response to methotrexate or other conventional DMARDs
- Moderate to severe disease
- Need for more targeted immunomodulation

Examples:

Drug	Mechanism
Adalimumab, Etanercept	TNF- α inhibitors
Abatacept	T-cell inhibition
Rituximab	B-cell depletion
Tocilizumab	IL-6 inhibition

Adverse Effects:

- INFECTION
- Infusion reactions
- All biologics increase infection risk.
- TB and hepatitis screening are important before treatment.



Can we target signaling pathways directly?

TARGETED SYNTHETIC DMARDS

Oral alternative to biologics

Oral small-molecule agents that inhibit intracellular signaling pathways

JAK-STAT pathway inhibition

Targeted synthetic
AK inhibitors (Tofacitinib, Baricitinib)

JAK Inhibitors

- These are oral targeted synthetic DMARDs
- They inhibit JAK-STAT signaling downstream of cytokine receptors
- Used selectively (**RESTRICTED**) due to safety concerns

MOA:

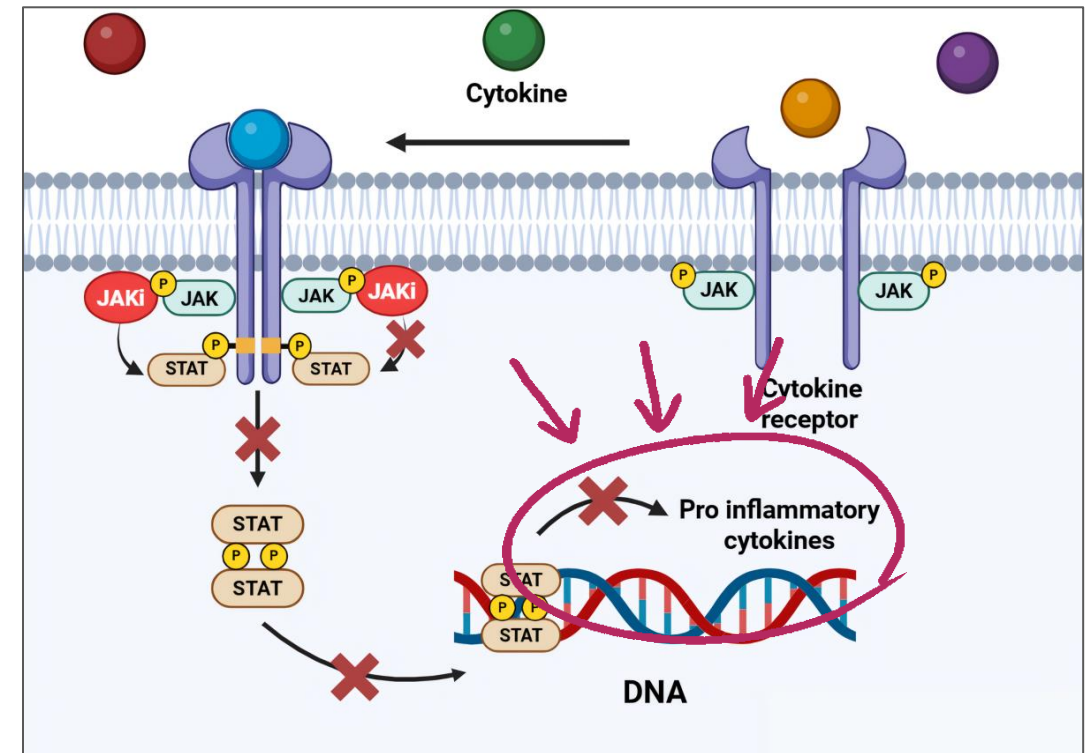
Block intracellular cytokine signaling

Examples: Tofacitinib, Baricitinib

Adverse Effects:

- **INFECTIONS, CVS events**
- Herpes zoster/ TB reactivation
- Cytopenias
- Lipid elevation
- **THROMBOSIS** warning

For illustration only



JAK/STAT pathway driving **inflammation**, **pannus formation**, and **joint destruction**. It acts as an intracellular signaling cascade for **pro-inflammatory cytokines** that are elevated in RA. Inhibiting this pathway **reduce inflammation** and improve clinical symptoms

CLINICAL INTEGRATION



NSAIDs vs. Steroids vs. DMARDs

Drug	Onset	Function	Disease progression
NSAIDs	Fast	Symptom relief	No
Steroids	Fast	Strong anti-inflammatory + immunosuppressive	Limited
DMARDs	Slow	Disease control	Yes

Steroids used as bridge therapy

- In RA, start MTX + prednisolone
- NSAIDs → analgesic/anti-inflammatory (symptomatic only)
- Steroids → disease-modifying bridge (temporary immunosuppression)

Bridge therapy means using short-term glucocorticoids (steroids) to rapidly control inflammation while waiting for DMARDs to start working (DMARDs takes weeks to show effects)

➤ **NSAIDs and steroids do not replace DMARDs in disease control.**

DMARDs Safety Measures

Before starting therapy
Infection history
Pregnancy considerations
Baseline CBC, LFT, renal function
TB and hepatitis screening where relevant, especially before biologics and JAK inhibitors

During therapy	
Periodic lab monitoring	
Infection surveillance	
Toxicity surveillance based on the specific drug	
Methotrexate	CBC, LFT
Hydroxychloroquine	Eye exam
Leflunomide	LFT
TNF inhibitors	TB screening
JAK inhibitors	CBC , lipids

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